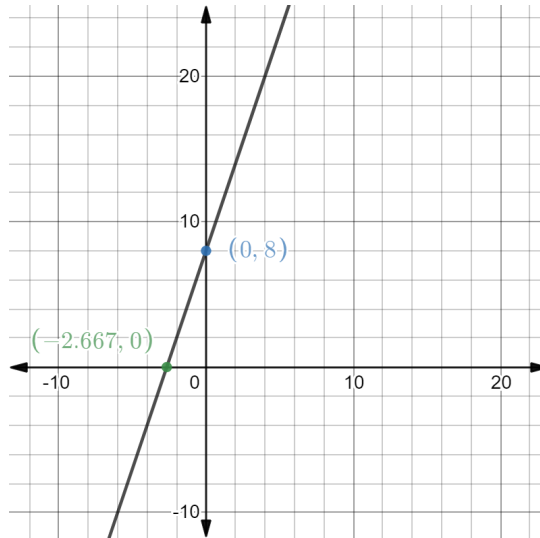




4.4.4 Wrap-Up: Ticket Out the Door

Antonella places x marbles into an empty container. Each marble has the same weight.

The weight, W , in ounces, of the container and the marbles can be found using the equation $W(x) = 3x + 8$.



- Li Qiang completed the table of four key features. Determine if the key features given by Li Qiang are viable. If the given key feature is not viable for the context, explain why it is NOT viable.

	Domain	Range	x -intercept	y -intercept
Key Feature	$x \geq 0$	$y \geq 0$	$(-\frac{8}{3}, 0)$	$(0, 8)$
Viable? Yes/No				
Explanation				



Unit 4, Lesson 5: Graphing and Interpreting Key Features of Lines



Benchmark

MA.912.AR.2.4 Given a table, equation, or written description of a linear function, graph that function, and determine and interpret its key features.



Learning Target

- ☐ I can graph a line and interpret its key features.



4.5.1 Warm-Up: Counterexamples

A counterexample is a statement that shows that a general statement is false. For example, a counterexample for the general statement “All birds can fly” is “No, penguins are birds and they cannot fly.”

1. Choose three of the statements and write a counterexample for each.

Statement	Counterexample
All prime numbers are odd.	
Division of two values always results in a smaller value.	
The product of two numbers is greater than their sum.	
All lines either increase from left to right or decrease from left to right.	
All lines have a y -intercept.	



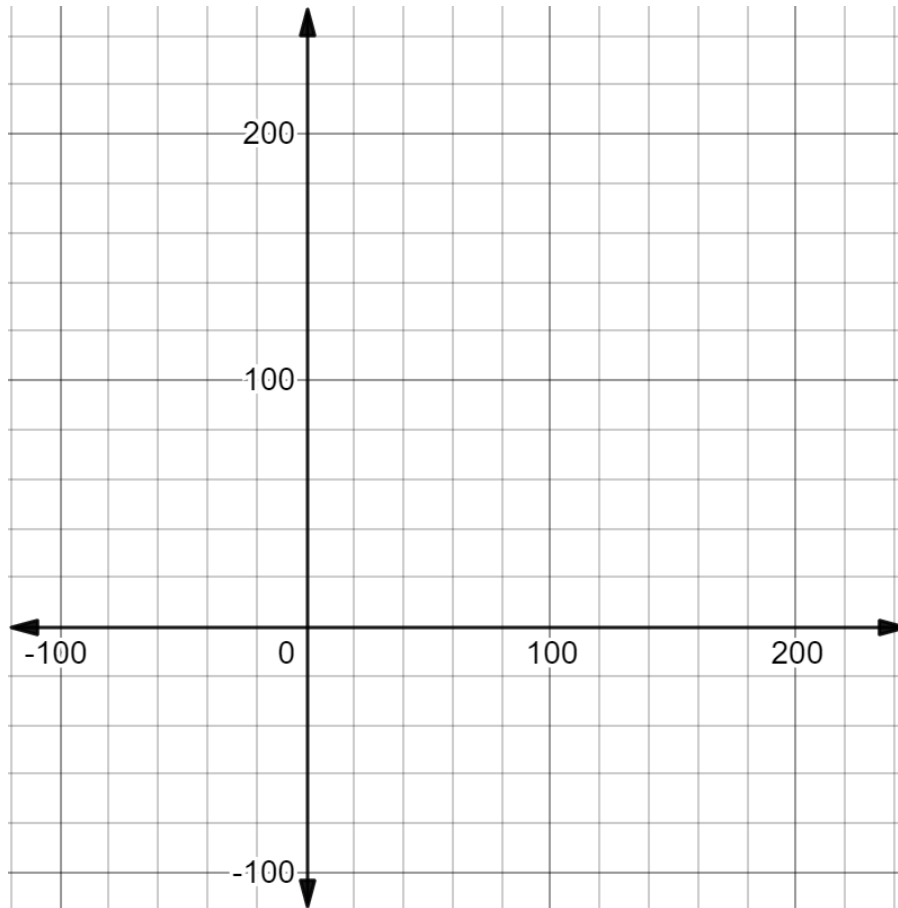
4.5.2 Guided Instruction: Graphing a Line and Interpreting Key Features

1. Lake Wales Little Theater is an intimate 140-seat theater. The theater uses the equation $16x + 8y = 1120$ to ensure each performance earns \$1,120 to cover the costs for a children’s production. A table of values is shown.

Adult Tickets x	Child Tickets y
0	140
18	104
25	90
56	28
70	0

- a. Rewrite $16x + 8y = 1120$ in slope-intercept form.

b. Graph the equation $16x + 8y = 1120$.



c. Circle the x -intercept in the table and place a point on the graph.

d. Complete the interpretation of the x -intercept.

Lake Wales Little Theater will meet production costs

when it sells

- ☐ 0
- ☐ 70
- ☐ 140

- ☐ adult
- ☐ child

tickets.

e. Circle the y -intercept in the table and place a point on the graph.

f. Complete the interpretation of the y -intercept.

Lake Wales Little Theater will meet production costs

when it sells

- ☐ 0
- ☐ 70
- ☐ 140

- ☐ adult
- ☐ child

tickets.

g. Complete the interpretation of the slope.

For every _____ child ticket(s) sold, Lake Wales

Little Theater needs to sell _____ fewer adult ticket(s).

h. Complete the table that gives the other key features. Also write a brief explanation for each.

	Domain	Range	As x -values increase, y -values ...
Key Feature			
Reason			

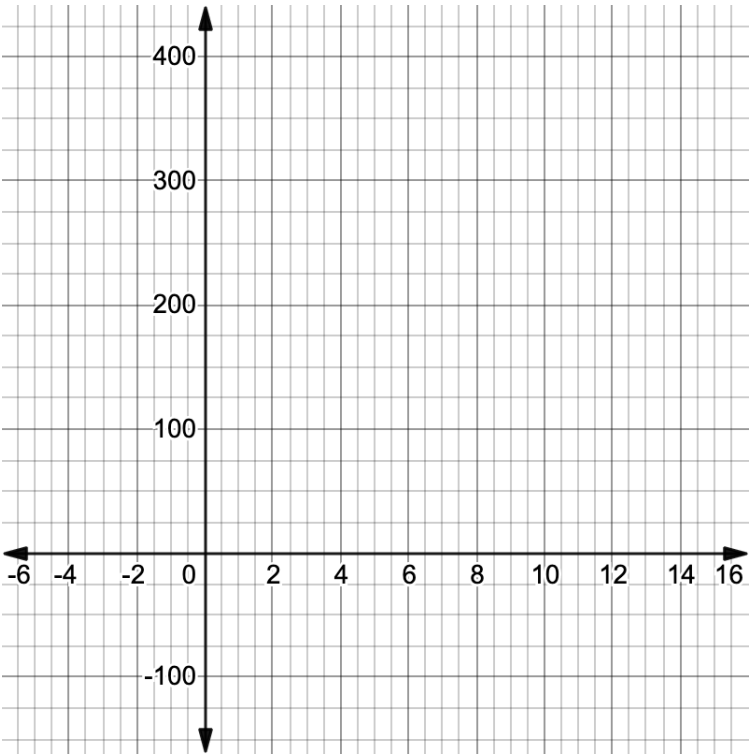


4.5.3 Try It: Graphing and Interpreting Key Features

The monarch butterfly is essential to the pollination process of many native and farmed North American plants. These butterflies migrate thousands of miles over several months.

A Florida Department of Forestry scientist has been tracking their population since the 2005 migration cycle. The scientist uses the function $B(x) = -35x + 325$ to estimate the number of butterflies, where x is the number of years since 2005 and B is the number of butterflies, in thousands.

1. Graph the function.



- 2. Solve $-35x + 325 = 0$.
- 3. Which intercept is the solution to $-35x + 325 = 0$?
- 4. Interpret the intercept.
- 5. Find $B(0)$.
- 6. Which intercept is the value of $B(0)$?
- 7. Interpret the intercept.

8. Complete the table.

	Domain	Range	As x -values increase, y -values ...
Key Feature			
Reason			

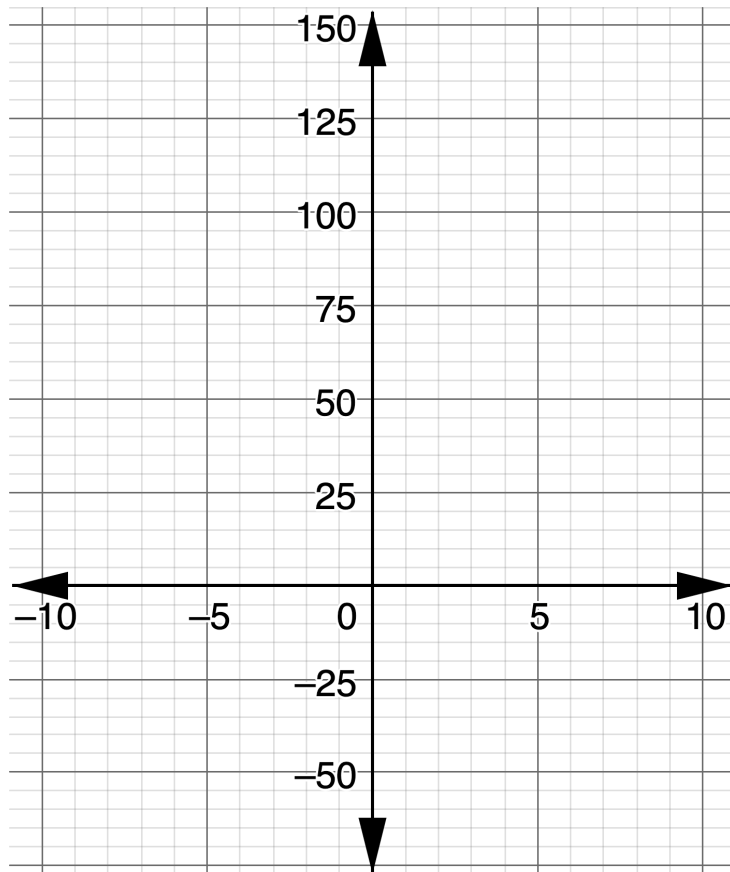




4.5.4 Your Turn!

At Pensacola Fitness Gym, the cost for a gym membership is given by the equation $y = 32.50x + 65$, where y is the total cost in dollars for a membership for a certain number of months, x . The gym charges an initial membership fee and a monthly fee.

1. Graph the equation.



2. Determine the key features.

Domain	Range	As x -values increase, y -values ...
Slope	x -intercept	y -intercept

3. Interpret the y -intercept.

4. Interpret the slope.



4.5.5 Wrap-Up: Error Analysis

The key features and the interpretations given for the context have errors. Find one of the errors and fix it.

- The average price of a gallon of orange juice from October 2013 to September 2014 can be modeled by the function f shown, where x represents the number of months since October 2013.

$$f(x) = 6.12 + 0.03x$$

	Domain	Range	x -intercept
Key Feature	$x \geq 0$	$y \geq 6.12$	6.12

Interpretations	
Slope	y -intercept
For every gallon of orange juice, the month increased by 0.03.	In October 2013, the average price of a gallon of orange juice was \$6.12.



Unit 4, Lesson 6: Determining Constraints



Benchmarks

MA.912.AR.2.5 Solve and graph mathematical and real-world problems that are modeled with linear functions. Interpret key features and determine constraints in terms of the context.

MA.912.F.1.2 Given a function represented in function notation, evaluate the function for an input in its domain. For a real-world context, interpret the output.



Learning Targets

- ☐ I can graph a line and interpret key features.
- ☐ I can determine constraints in terms of the context.
- ☐ I can evaluate and interpret function notation in terms of context.